4.33

(a)

$$P(X < 78) = P\left(Z < \frac{78 - 60}{30}\right) = \Phi(0.6) = 0.7257$$

(b) $0.88 = \Phi(z) \Rightarrow z = 1.175 \Rightarrow x = \mu + \sigma z = 60 + 30(1.175) = 95.25$

4.39

(a)

$$P(X < 85) = P\left(Z < \frac{85 - 100}{15}\right) = \Phi(-1) = 0.1587$$

(b) $0.75 = \Phi(z) \Rightarrow z = 0.67 \Rightarrow x = \mu + \sigma z = 100 + 15(0.67) = 110$

(c)

$$P(X > 120) = P\left(Z > \frac{120 - 100}{15}\right) = P(Z > 1.3333) = 1 - \Phi(1.3333) = 1 - 0.9088$$
$$= 0.0912$$

4.43

(a) $0.7 = \Phi(z) \Rightarrow z = 0.52 \Rightarrow x = \mu + \sigma z = 39 + 0.52 = 39.52$

$$P(X < 39.52) = P\left(Z < \frac{39.52 - 40}{1}\right) = \Phi(-0.48) = 0.32$$

32nd percentile

(b)

$$P(37 < X < 40) = P\left(\frac{37 - 39}{1} < Z < \frac{40 - 39}{1}\right) = P(-2 < Z < 1)$$
$$= \Phi(1) - \Phi(-2) = 0.8413 - 0.0228 = 0.8185$$

4.45

$0.77 = \Phi(z) \Rightarrow z = 0.74 \Rightarrow x = \mu + \sigma z = 70 + 12(0.74) = 78.88$

The lowest possible score for an A is 79.

4.50

(a)

```
>pbinom(65, 150, 0.4)-pbinom(47, 150, 0.4)
0.8029938
>1-pbinom(69, 150, 0.4)
0.05745956
>pbinom(66, 150, 0.4)-pbinom(48, 150, 0.4)
0.8340038
>1-pbinom(64, 150, 0.4)
0.2259444
>pbinom(59, 150, 0.4)
0.4690135
```

(b)

```
>pnorm(65.5, 60, 6)-pnorm(47.5, 60, 6)
0.8017309
>1-pnorm(69.5, 60, 6)
0.05667275
>pnorm(66.5, 60, 6)-pnorm(48.5, 60, 6)
0.8330296
>1-pnorm(64.5, 60, 6)
0.2266274
>pnorm(59.5, 60, 6)
0.4667932
```

4.53

Let X be the number of people who have type O blood. Then $X \sim \mathrm{Bin}(50, 0.45)$.

$\mu = np = 50(0.45) = 22.5, \qquad n(1-p) = 50(0.55) = 27.5$ are large.

$$\sigma = \sqrt{np(1-p)} = \sqrt{50(0.45)(0.55)} = 3.52$$

$$P(X > 25) \approx P\left(Z > \frac{25.5 - 22.5}{3.52}\right) = P(Z > 0.85) = 1 - \Phi(0.85) = 1 - 0.8023$$
$$= 0.1977$$

4.55

Let X be the number of heads. Then $X \sim \text{Bin}(200, 0.5)$.

$\mu = np = 200(0.5) = 100,\ n(1-p) = 200(0.5) = 100$ are large.

$$\sigma = \sqrt{np(1-p)} = \sqrt{200(0.5)(0.5)} = 7.071$$

(a)

$$P(X < 90) \approx P\left(Z < \frac{89.5 - 100}{7.071}\right) = \Phi(-1.485) = 0.0688$$

(b)

$$\begin{aligned} P(X = 100) &\approx P\left(\frac{99.5 - 100}{7.071} < Z < \frac{100.5 - 100}{7.071}\right) \\ &= P(-0.0707 < Z < 0.0707) = \Phi(0.0707) - \Phi(-0.0707) \\ &= 0.5282 - 0.4718 = 0.0564 \end{aligned}$$

4.61

(a)

$$f(x) = \begin{cases} \dfrac{\Gamma(5)}{\Gamma(4)\Gamma(1)} x^{4-1}(1-x)^{1-1} = 4x^3, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

(b)

$$F(x) = \begin{cases} 0, & x \le 0 \\ \displaystyle\int_{-\infty}^{x} f(t)dt = \int_0^x 4t^3 dt = t^4|_0^x = x^4, & 0 < x < 1 \\ 1, & x \ge 1 \end{cases}$$

(c)

$$P(0.25 < X < 0.75) = F\left(\frac{3}{4}\right) - F\left(\frac{1}{4}\right) = \frac{81}{256} - \frac{1}{256} = \frac{80}{256} = \frac{5}{16} = 0.3125$$

(d)

$$E(X) = \frac{\alpha}{\alpha + \beta} = \frac{4}{5}$$

$$Var(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)} = \frac{4}{5^2 \cdot 6} = \frac{4}{150} = \frac{2}{75}$$

(e)

$$\frac{1}{2} = F(x) = x^4 \Rightarrow x = \sqrt[4]{1/2} = 0.8409$$

4.63

(a)

$$f(x) = \begin{cases} \frac{\Gamma(5)}{\Gamma(2)\Gamma(3)} x^{2-1}(1-x)^{3-1} = \frac{4!}{2!} x(1-x)^2 = 12(x - 2x^2 + x^3), & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

(b) For $0 < x < 1$,

$$F(x) = \int_{-\infty}^{x} f(t)dt$$

$$= \int_{0}^{x} 12(t - 2t^2 + t^3)dt = (6t^2 - 8t^3 + 3t^4)|_0^x = 3x^4 - 8x^3 + 6x^2$$

Hence

$$F(x) = \begin{cases} 0, & x < 0 \\ 3x^4 - 8x^3 + 6x^2, & 0 \le x \le 1 \\ 1, & x > 1 \end{cases}$$

(c)

$$E(X) = \frac{\alpha}{\alpha + \beta} = \frac{2}{5}$$

(d)

$$P(X \ge 0.2) = 1 - F(0.2) = 1 - (3(0.2)^4 - 8(0.2)^3 + 6(0.2)^2) = 0.8192$$

(e)

$$\begin{aligned} P(0.1 < X < 0.25) &= F(0.25) - F(0.1) \\ &= (3(0.25)^4 - 8(0.25)^3 + 6(0.25)^2) \\ &\quad - (3(0.1)^4 - 8(0.1)^3 + 6(0.1)^2) = 0.2094 \end{aligned}$$

4.65

(a) Answer: 1

$$\Phi(2) + \Phi(-2) = \Phi(2) + [1 - \Phi(2)] = 1$$

(b) Answer: 0.75

pnorm(1,1,4) is the command for $P(X < 1)$, where $X \sim N(1, 4^2)$.
Thus, the output of Pnorm(1,1,4) is 0.5.
punif(0.5,-1,1) is the command for $P(Y < 0.5)$, where $Y \sim \text{Unif}(-1, 1)$.
Hence, the output of punif(0.5,-1,1) is $P(Y < 0.5) = 0.75$.

(c) Answer: 3

Since qgamma is the inverse of pgamma, the answer is 3.

(d) Answer: NaN (error)

Since qexp is a command for finding a percentile, the input should be between 0 and 1.